

Supplementary File for:
Study of Cis-regulatory Effects at the Population
Level

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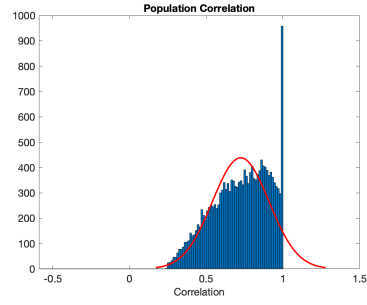
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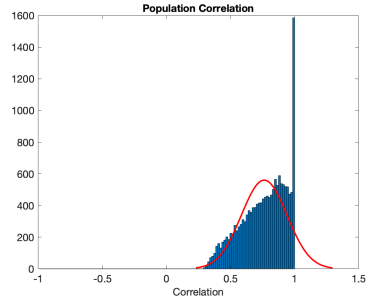
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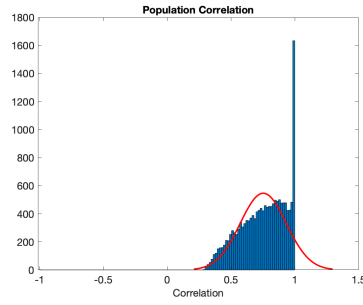
Figures



(a) Leaves



(b) Berries Soft



(c) Berries Hard

Figure S1: distribution of the statistical significant correlations between the values of AEI estimated by using EGC and the ones expected for that AEI using the RNA-seq read counts used in our experiments.

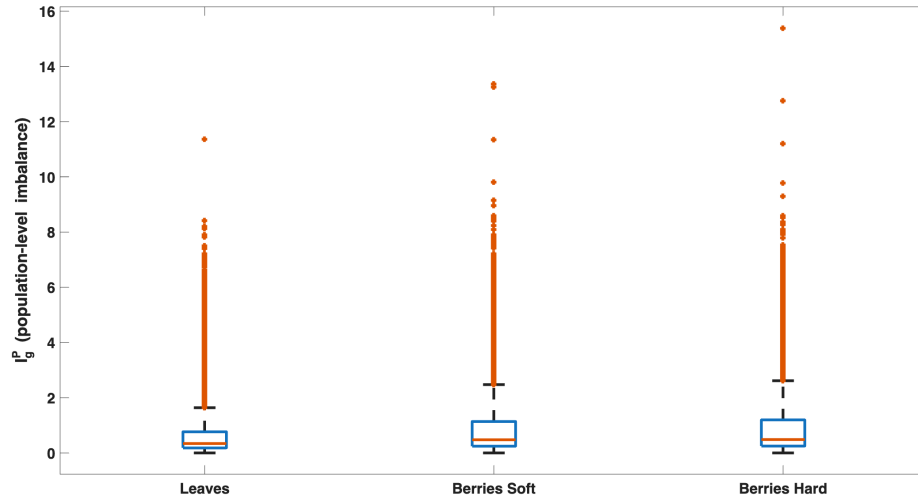


Figure S2: log10-transformed distribution of $I_{\mathcal{P}}^g$ across tissues. Boxplots show increasing dispersion and higher-value outliers from leaves to berries, indicating stronger and more heterogeneous cis-regulatory effects in fruit tissues.

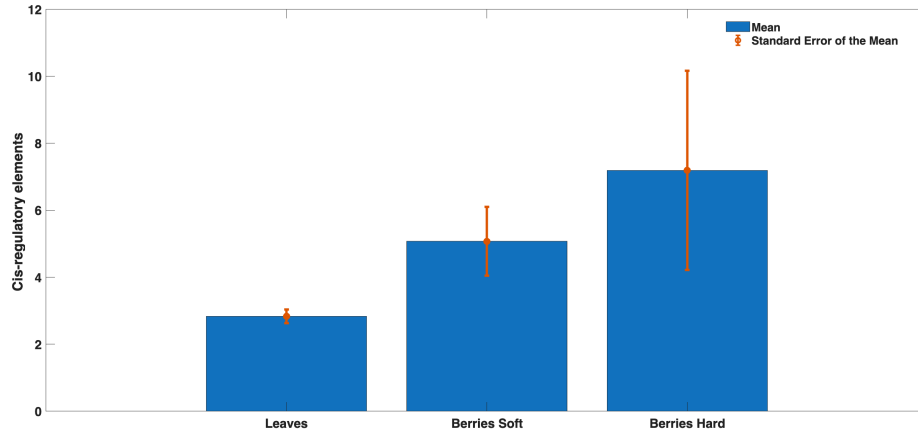


Figure S3: mean value of statistical significant $I_{\mathcal{P}}^g$ across tissues. Error bars represent the standard error of the mean (SEM). The increasing trend from leaves to berries indicates stronger cis-regulatory imbalance in fruit tissues.

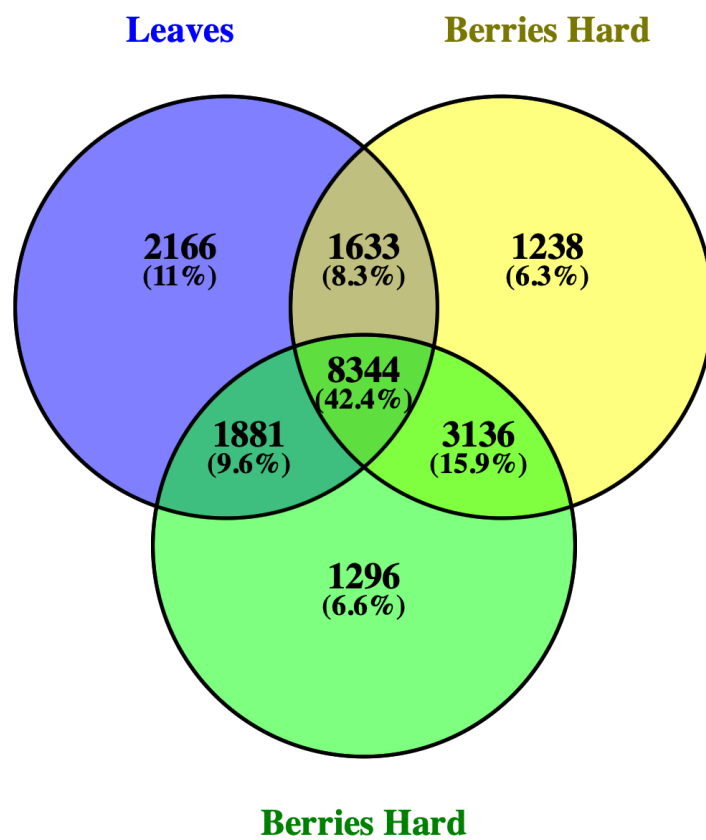


Figure S4: intersections between sets of genes having statistical significant ($FDR \leq 0.05$) imbalance of cis-regulation in the different tissues under the FUM model.

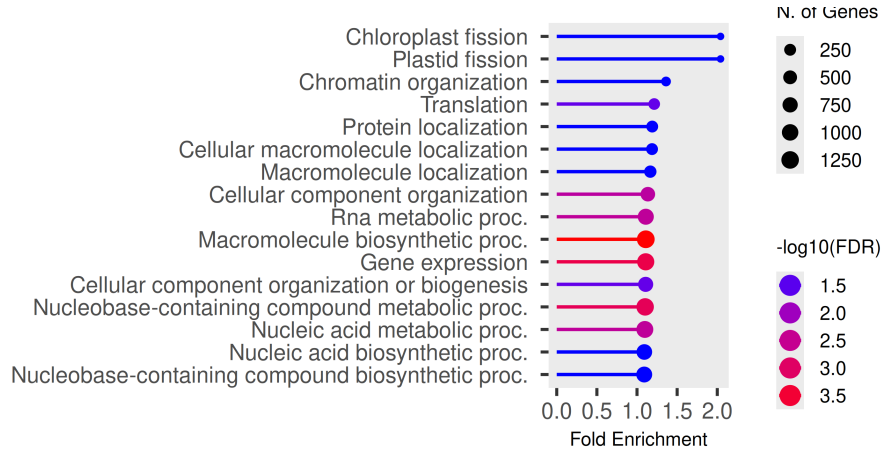


Figure S5: Gene Ontology (GO) enrichment analysis of the the genes having statistical significant I_P^g ($FDR \leq 0.05$) that are shared among the three tissues. The plot shows significantly enriched ($FDR \leq 0.05$) GO Biological Process terms for the input gene list. The x-axis indicates fold enrichment relative to the genomic background, obtained by considering all the genes of the grapevine variants dataset, while line colors represent statistical significance expressed as $-\log_{10}(\text{FDR})$. FDRs have been computed by applying the Benjamini-Hochberg method to p-values computed by means of hypergeometric test. Fold enrichment is defined as the percentage of genes in your list that are in a pathway/gene set divided by the corresponding percentage in the background genes. While FDR measures statistical significance, fold enrichment indicates effect size. Pathways and gene sets have been obtained from [1].

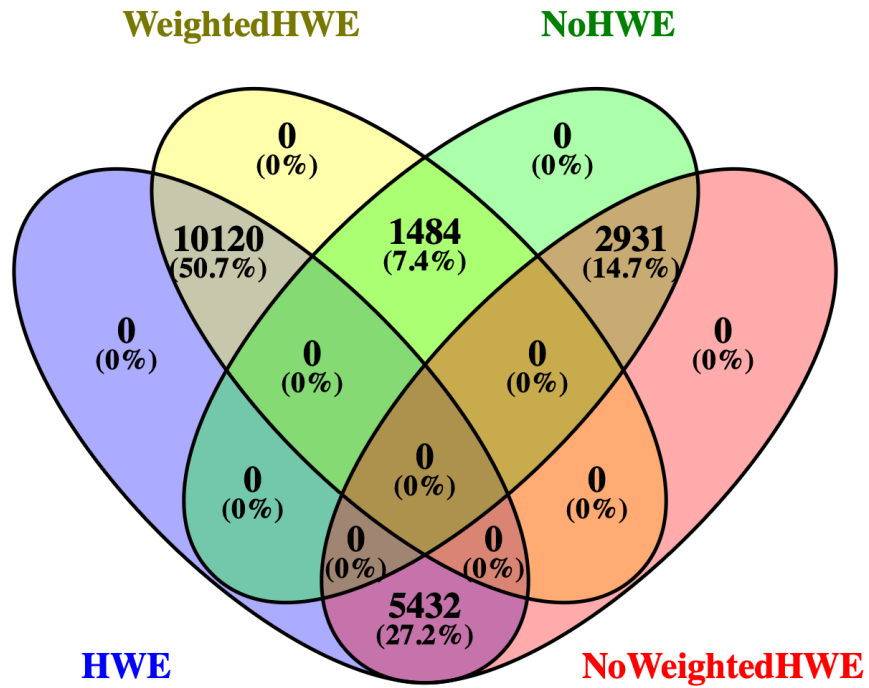


Figure S6: Venn diagram that compares the results for leaves of HWE and weighted HWE tests discussed in Figure S6. HWE = set of genes that are in equilibrium according to HWE test; NoHWE = set of genes that are not in equilibrium according to HWE test; WeightedHWE = set of genes that are in equilibrium according to weighted HWE test; NoWeightedHWE = set of genes that are not in equilibrium according to weighted HWE test.

Abbreviations

Symbol	Description
g	Gene under analysis.
$\mathcal{A}_i^g$	i -th allele of gene g .
E_k^g	Total expression level of gene g in individual k .
$E_k^{A_{g_i}^g}$	Expression contribution of allele $A_{g_i}^g$ in individual k .
C_k^g	Cis-regulatory contribution to expression.
T_K^G	Trans-acting contribution to expression.
ϵ_k	Residual noise term.
e_i^g	Estimated Genetic Contribution (EGC) of allele $\mathcal{A}_i^g$.
I_k^g	Allelic imbalance of gene g in individual k .
$I_{\mathcal{P}}^g$	Population-level imbalance statistic.
$p_i^g(t)$	Frequency of allele $\mathcal{A}_i^g$ at time t .
$p_{i,j}^g(t)$	Frequency of genotype $\mathcal{A}_i^g \mathcal{A}_j^g$ at time t .
$n_{i,j}^g$	Imbalance between alleles $\mathcal{A}_i^g$ and $\mathcal{A}_j^g$.
$\tilde{p}_i^g(t)$	Weighted allele frequency based on EGC.
$\mathcal{P}$	Population of individuals.
m	Population size.
$S_{I_{\mathcal{P}_s}^g}$	Set of simulated imbalance statistics.

Table 1: Summary of notation used in the manuscript.

References

- [1] Xiaoya Shi, Shuo Cao, Xu Wang, Siyang Huang, Yue Wang, Zhongjie Liu, Wenwen Liu, Xiangpeng Leng, Yanling Peng, Nan Wang, Yiwen Wang, Zhiyao Ma, Xiaodong Xu, Fan Zhang, Hui Xue, Haixia Zhong, Yi Wang, Kekun Zhang, Amandine Velt, Komlan Avia, Daniela Holtgräwe, Jérôme Grimplet, José Tomás Matus, Doreen Ware, Xinyu Wu, Haibo Wang, Chonghuai Liu, Yuling Fang, Camille Rustenholz, Zongming Cheng, Hua Xiao, and Yongfeng Zhou. The complete reference genome for grapevine (*Vitis vinifera* L.) genetics and breeding. *Horticulture Research*, 10(5):uhad061, 04 2023.